Technical University of Munich

Network Compression Bearing No Trade-Offs on Qubits

Tigran Bunarjyan

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Supervisors: Dai Liu, Xiaorang Guo
Examiner: Prof. Dr. Martin Schulz

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ABSTRACT

In superconducting quantum computing, achieving high-speed, high-accuracy qubit readout is essential for reliable mid-circuit measurements and fault-tolerant computation. This research primarily focuses on network compression techniques for qubit readout models, aiming to reduce computational demands without significant performance degradation. Traditional deep learning-based readout methods, while effective, typically rely on synchronized multi-qubit discrimination, which introduces challenges in adaptability and timing—especially in scenarios requiring mid-circuit measurements.

To facilitate efficient compression, we first design independent neural network discriminators for each qubit, thereby eliminating the need for synchronized outputs and allowing immediate, high-fidelity discrimination of each qubit individually. This approach supports real-time feedback and enhances system responsiveness, making it well-suited for practical quantum applications.

Our research incorporates well-known advanced compression methods, including knwoledge distillation, pruning, and quantization, to come up with lightweight, hardware-efficient models that retain high accuracy. Extensive experiments demonstrate that our compressed models achieve accuracy levels comparable to their uncompressed counterparts, with significant reductions in memory and computational requirements. This makes them practical for FPGA-based systems and other resource-constrained hardware. The resulting architecture thus provides a scalable, efficient solution for qubit readout in quantum processors, supporting the implementation of real-time adaptive algorithms and error-corrected quantum computing applications.

1 Introduction and Motivation

Quantum computing promises substantial computational advantages over classical computing, particularly in solving complex problems. However, realizing these advantages hinges on the development of large-scale, high-fidelity quantum devices, and, more specifically, on the implementation of robust and efficient qubit readout methods as quantum processors scale from tens to thousands of qubits.

The challenge of qubit readout—accurately and rapidly measuring qubit states in quantum computing—has become a critical area of research due to its central role in quantum information processing. Effective qubit readout mechanisms are essential not only for reliable computation but also for quantum error correction and feedback control, especially given the iterative and repeated measurements required in many quantum algorithms.

Superconducting qubits, one of the leading quantum computing platforms, have shown remarkable progress, including coherence times exceeding 100 microseconds and high-fidelity gate operations. Despite these advancements, achieving high-fidelity readout, particularly in multi-qubit systems, remains a major challenge due to factors such as crosstalk and noise interference. Recent approaches utilizing deep neural networks (DNNs) for multi-qubit state discrimination have demonstrated promising results, with Blienhart et al. (2021) [1] achieving notable error reduction using feedforward neural networks (FNNs) that mitigate multi-qubit interference through machine learning.

Nevertheless, multi-qubit systems continue to face readout delays and computational complexity that impact the efficiency of quantum operations. While simultaneous readout using a single model for multiple qubits streamlines computation, it also presents limitations: these models require all qubit data to be available before state discrimination, introducing latency and limiting applicability in real-time quantum systems. Addressing these constraints is essential to realize scalable, high-performance quantum computing.

Our research specifically addresses the following questions:

- **Is it possible to achieve high-fidelity qubit readout with independent neural networks for single-qubit discrimination, while maintaining comparable performance to synchronous multi-qubit models?**

- **Can network compression techniques, such as knwolede distillation, network pruning and quantization, reduce computational complexity without compromising readout accuracy?**

To address these questions, we propose a readout method that supports independent state discrimination for each qubit, thereby reducing latency by allowing each qubit to be processed as soon as it is ready. This approach circumvents the synchronization limitations of collective qubit readout, which are incompatible with mid-circuit measurements required in error-corrected and adaptive quantum algorithms. Additionally, to ensure computational efficiency and facilitate deployment on resource-constrained hardware, such as FPGAs, we explore network compression techniques. By leveraging knwoledge distillation and other methods, our goal is to develop a streamlined, high-accuracy readout structure that supports scalable quantum processing and facilitates practical implementation in quantum control systems.

Our research explores two strategic approaches to achieving these goals:

1. **Implementing independent single-qubit discriminators:** Structuring the readout task into individual single-qubit discriminators eliminates the need for collective data readiness, enabling immediate assessment of each qubit's state. This approach reduces readout latency, minimizes crosstalk complexity, and offers a scalable solution for larger quantum systems where simultaneous multi-qubit discrimination becomes computationally prohibitive.

2. **Applying compression techniques to reduce computational complexity:** Through network compression methods, such as network pruning, knowledge distillation and quantization, we aim to reduce the computational burden while maintaining high accuracy. This approach enhances computational efficiency by lowering memory and power demands, enabling the deployment of qubit discriminators on specialized hardware, even in resource-limited environments.

Our investigation into network compression for single-qubit discriminators is designed to balance accuracy and efficiency, retaining geometric mean accuracy comparable to that of uncompressed models. This approach not only preserves essential readout fidelity but also broadens applicability, allowing compressed models to be effectively integrated within edge quantum processors, including those operating within constrained hardware systems.

2 Related Art

Application of deep learning for qubit readout has gained significant attention due to its potential to improve measurement fidelity and efficiency in quantum systems. Blienhart et al. [1] first demonstrated the application of deep neural networks (DNNs) for multi-qubit state discrimination in superconducting systems, using feedforward neural networks (FNNs) to mitigate interference and crosstalk between qubits, which enhanced readout fidelity. However, their reliance on synchronized qubit data introduced latency, limiting scalability for real-time applications like quantum error correction.

Other researchers explored recurrent neural networks (RNNs) and convolutional neural networks (CNNs) to capture temporal dependencies in qubit signals, improving accuracy in noisy environment in Krastanov et al. [2]. Additionally, Zhang et al. [3] applied variational quantum classifiers to enhance discrimination accuracy in larger qubit systems. Similarly, Magesan et al. [4] employed machine learning techniques to enhance readout fidelity by correcting measurement errors, though these approaches did not address the computational challenges and latency associated with multi-qubit systems requiring independent, real-time data processing.

A key limitation of existing deep learning-based methods is the need for synchronized multi-qubit data, which leads to high computational demands and latency. These models typically rely on large neural networks, making them unsuitable for hardware-constrained environments like FPGAs. This challenge has driven interest in optimizing neural networks for quantum readout through compression techniques. Recent studies by Wang et al. [5], Shen et al. [6], and Han et al. [7] have explored knowledge distillation, pruning, and quantization to compress neural networks, although their application to qubit readout remains limited. These methods, successful in classical machine learning [8], are promising for reducing computational and memory requirements in quantum environments with limited hardware resources.

In our work, we explore the potential of deep learning for independent qubit readout, aiming to reduce synchronization delays and latency. By experimenting with network compression techniques, we seek to develop lightweight, hardware-efficient models suitable for FPGA-based platforms. While existing research does not combine independent qubit discrimination with these compression techniques, our goal is to validate whether this combination can offer a scalable solution for qubit readout and address some limitations of traditional multi-qubit models, particularly in terms of scalability and hardware constraints. We hope that this approach will support real-time quantum measurements, enabling more efficient adaptive quantum algorithms and error-corrected systems. Preliminary results suggest that the compressed architecture achieves performance comparable to uncompressed models while significantly reducing memory and computational demands, making it potentially beneficial for resource-constrained environments.

3 Superconducting Qubit Readout Data

This study utilizes data generated from a superconducting quantum computing setup featuring five frequency-tunable transmon qubits arranged in a linear array. Each qubit operates in a frequency range between 4.3 GHz and 5.2 GHz and is read out through a multiplexed, frequency-tuned approach [1]. Specifically, the dataset comprises time-series recordings of the in-phase (I) and quadrature (Q) components of the readout signal, essential for accurately determining each qubit's state.

3.1 Data Collection and Measurement

The I/Q data are obtained through dispersive readout in a circuit quantum electrodynamics (cQED) setup, where each qubit couples to a detuned resonator. This coupling induces a qubit-state-dependent frequency shift, which manifests as a phase shift in the I/Q signals upon transmission or reflection of a coherent microwave signal. Consequently, the I/Q data carry information about each qubit's state, but are also influenced by noise and crosstalk from neighboring qubits due to the shared readout architecture [1].

Data collection involves transmitting a frequency-multiplexed readout tone through a shared feedline that connects all five qubits. Each qubit is linked to this feedline via a co-planar waveguide resonator, with Purcell filters integrated to minimize off-resonant energy decay [1]. The readout tone consists of superposed baseband signals at intermediate frequencies (IF) between 10 MHz and 150 MHz, which are then upconverted to each qubit's resonant readout frequency. As the signal propagates through the feedline, it undergoes down-conversion using a shared local oscillator, followed by digitization of the I/Q components at a 2 ns sampling period. This setup is enhanced with a Josephson traveling-wave parametric amplifier (JTWPA) at 20 mK and a high-electron-mobility transistor (HEMT) amplifier at 3 K to improve the signal-to-noise ratio, thereby enabling precise qubit state discrimination.

The I/Q data play a critical role in training machine learning models for accurate qubit state discrimination. However, frequency-multiplexed, shared readout introduces challenges such as crosstalk and other complex error sources, which complicate state discrimination. Crosstalk errors, for example, may arise from qubit interactions or limited spectral separation between readout frequencies, increasing the complexity of the I/Q signal data and necessitating robust machine learning techniques for effective qubit state classification [1].

Given these data characteristics, advanced deep learning methods are needed to effectively classify qubit states from noisy, multiplexed signals. This dataset, originally described in Blienhart et al. (2021) [1], provides a realistic and complex testing ground for evaluating the performance of the independent qubit discriminator models and compression techniques developed in this study.

3.2 Dataset Description

The dataset used in this study consists of both training and testing data, structured to capture time-series measurements for qubit state discrimination. The training data have a shape of (480,000, 1000, 2), where 480,000 represents the total number of samples. Each sample spans a trace length of 1000 time bins resulting in a 2-microsecond measurement duration per sample. Similarly, the testing dataset has a shape of (1,120,000, 1000, 2), providing an expanded set for model evaluation. For both datasets, the final dimension of 2 corresponds to the I (in-phase) and Q (quadrature) components, which together encode the qubit state information. These data capture measurements across five qubits arranged in the order [QB5, QB4, QB3, QB2, QB1], facilitating comprehensive analysis and model training in the context of qubit readout discrimination.

The dataset includes labels representing all possible combinations of qubit states for the five-qubit system, ranging from `00000` to `11111`, where each bit corresponds to the state of an individual qubit. With five qubits, there are $2^5 = 32$ possible state combinations, each serving as a unique class label. Each label in the dataset indicates the combined state of all qubits for a given sample, with `0` and `1` representing the ground and excited states of each qubit, respectively. Thus, the original dataset's labels can enable models to distinguish between these 32 unique classes, allowing for precise multi-qubit state discrimination.

3.3 Individual Qubit Data Processing

To conduct experiments focused on single-qubit readout, the original dataset was processed to create qubit-specific datasets for each of the five qubits. For each qubit i, a separate training and testing dataset was generated, consisting of binary labels that indicate whether the qubit is in the ground state ('0') or the excited state ('1').

The procedure for creating these qubit-specific datasets involved the following steps: for a given qubit i, rows from the original dataset were selected where the binary representation of the label had a '1' in the position corresponding to qubit i. These rows were assigned a positive class label ('1') for qubit i. All other rows, where qubit i was in the ground state, were assigned a negative class label ('0'). This process was repeated for each of the five qubits, yielding five separate datasets, each of equal size and containing binary labels specific to the respective qubit's state.

This data processing approach ensures that each qubit-specific dataset is balanced and can independently train and test models for single-qubit state discrimination, allowing for focused experimental analysis on the performance of each qubit's readout.

4 Single-qubit Discrimination

In this study, we focus on developing binary classifiers for single-qubit discrimination using deep learning architectures tailored for each qubit. Deep learning, particularly feedforward neural networks (FNNs), has proven highly effective for complex classification tasks by learning hierarchical representations from raw input data [9]. An FNN consists of multiple layers of neurons, each applying a transformation followed by an activation function. This network is trained by adjusting weights through backpropagation to minimize a loss function [10]. Here, we utilize this approach to distinguish between the ground and excited states of individual qubits based on I/Q signal data.

4.1 Model Architecture

After testing several architectures, we identified an optimal FNN structure that we named as `SingleQubitFNN`, which achieved the highest accuracy for single-qubit discrimination while maintaining efficiency across all five qubits when evaluated with geometric mean accuracy. The model architecture combines key components such as ReLU activations, dropout regularization, and batch normalization, which together ensure high stability and accuracy.

The `SingleQubitFNN` model processes I/Q time-series data for a single qubit with the following architecture:

- **Input Layer**: Accepts I/Q time-series data for a single qubit.
- **Hidden Layers** for 1μ trace duration input:
  - The neural network consists of three hidden layers, containing 500, 250, and 125 neurons, respectively.
- **Output Layer**: Outputs a binary prediction (0 or 1) for each qubit's state.

ReLU activations are applied after each hidden layer, followed by batch normalization to stabilize training and dropout (0.5 probability) to prevent overfitting.

4.2 Training and Evaluation

The model was trained on individual qubit datasets, with labels indicating whether each qubit was in the excited (1) or ground state (0). The training set consisted of 432,000 samples, with 48,000 for validation. We used a batch size of 128 and an initial learning rate of 0.0001, applying `BCEWithLogitsLoss` for binary classification [10].

The training process used binary-labeled data, with each qubit classified independently as being in the excited or ground state.

Loss Function, Optimizer, and Evaluation Metrics: We used the `BCEWithLogitsLoss` function, which combines binary cross-entropy with sigmoid activation, for binary classification tasks. The loss for a batch of N samples is computed as:

$$\text{Loss} = \frac{1}{N} \sum_{i=1}^{N} \left(\log(1 + e^{-z_i}) + (1 - y_i) \cdot z_i \right)$$

where z_i is the model output for the i-th sample, and y_i is the true label. The Adam optimizer was chosen for its efficiency, with a `ReduceLROnPlateau` scheduler applied to reduce the learning rate by 0.75 when the validation loss plateaued. Accuracy was the primary metric, with the geometric mean of individual accuracies calculated as:

$$GM = \left(\prod_{i=1}^{5} A_i \right)^{\frac{1}{5}}$$

where A_i denotes the accuracy of each qubit's classifier.

After training, models were evaluated on a test set of 1,120,000 samples. Accuracy was calculated for each qubit independently, with system-wide performance evaluated using the geometric mean to ensure balanced performance across all five qubits [11].

4.3 Experimental Results

We analyzed the effect of trace length (300, 500, 800, and 1000) on the model's accuracy. Longer trace lengths provided more data for the network, improving performance. Results from individual qubit models are summarized below.

Trace Duration	Qubit 1	Qubit 2	Qubit 3	Qubit 4	Qubit 5	Geom. Mean
0.6μ	0.953	0.727	0.937	0.936	0.969	0.899
1μ	0.969	0.746	0.940	0.946	0.970	0.910
1.6μ	0.971	0.757	0.941	0.947	0.970	0.913
2μ	0.971	0.761	0.940	0.946	0.970	0.914

Table 4.1: Classification Performance of Individual Qubit Discriminators for Various Trace Durations with Geometric Mean as an Aggregate Performance Metric for the Five Independent Neural Networks.

The experimental results demonstrate that FNNs are highly effective for single-qubit state discrimination. The individual qubit discriminator `SingleQubitFNN` model provided high accuracy for individual qubit classification, while the multi-input model showed potential for simultaneous multi-qubit state prediction with slightly higher accuracy [1] at longer trace lengths.

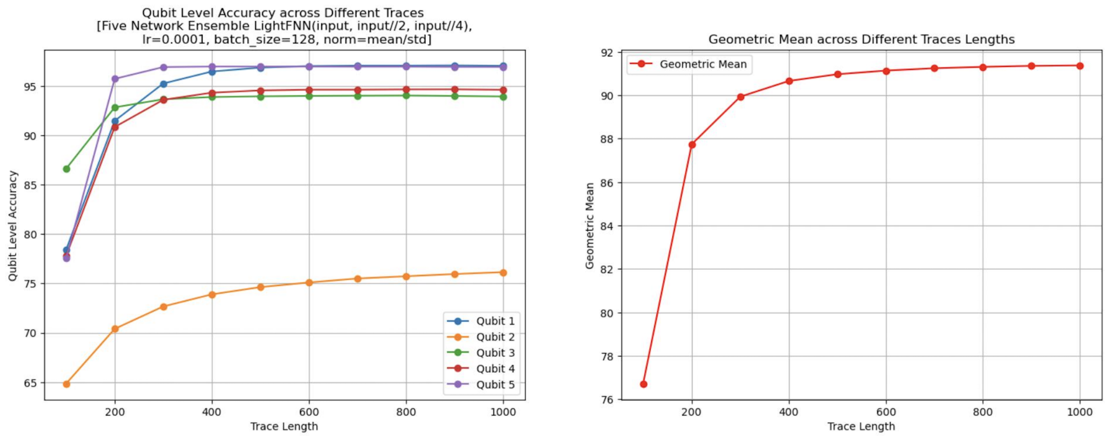

Figure 4.1: Visualization of the Accuracy of Five Individual Qubit Classifiers Across Various Trace Durations from 0.2μ to 2μ (left) and Geometric Mean Accuracy for the Five Neural Network System (right).

For the individual FNN classifiers applied to each qubit independently:

- **0.6** μ**s**: GM accuracy was 0.899, with notable variability—Qubit 2 had the lowest accuracy (0.727) and Qubit 5 the highest (0.969).
- **1** μ**s**: GM accuracy improved to 0.910, indicating that longer traces reduce misclassification.
- **1.6** μ**s**: GM accuracy further increased to 0.913, with Qubit 2 improving to 0.757.
- **2** μ**s**: GM accuracy peaked at 0.914, with only minor gains over 1.6 μs, suggesting diminishing returns.

The individual qubit readout models, referred to as Single Qubit NNs, demonstrated significant success in superconducting qubit state discrimination. This approach eliminates the need for simultaneous availability of all five qubits for processing, as required in previous methods such as in Blienhart et al.'s work [1]. By independently processing each qubit, latency is reduced, and computational efficiency is improved.

Additionally, optimal trace durations can be tailored for each qubit, further enhancing accuracy. This flexibility allows for adaptive tuning of the trace length per qubit, resulting in increased readout fidelity, particularly in scenarios with varying signal quality across qubits. The demonstrated performance across different trace lengths validates the efficacy of these FNN-based classifiers in handling qubit readout tasks in practical quantum computing applications.

5 Network Compression

As neural networks grow increasingly complex, particularly in quantum computing contexts where hardware efficiency is crucial, model compression techniques offer significant advantages. Compression reduces the size, memory footprint, and computational demands of a neural network while maintaining a comparable level of accuracy. Here we explore three prominent compression methods: knowledge distillation, network pruning, and quantization. These techniques enable deployment on resource-constrained devices, such as FPGA-based systems, by creating lightweight yet high-performing models.

5.1 Network Pruning for Qubit Discrimination

Network pruning reduces neural network parameters by removing redundant or non-critical connections, that can be applied at different levels:

- **Unstructured Pruning**: Removes individual weights, leading to sparse matrices, though this may not be hardware-efficient.
- **Structured Pruning**: Removes entire filters, neurons, or channels, maintaining a dense structure that is more suitable for hardware optimizations.
- **Global Pruning**: Removes connections based on their overall importance across the network, allowing pruning across layers, which offers more flexibility than layer-specific pruning.

Pruning is typically guided by a criterion $C(w)$, such as weight magnitude:

$$C(w) = |w|$$

Weights below a threshold are pruned, and the model is fine-tuned. In quantum computing, pruning reduces memory and computational load, improving inference times on resource-constrained hardware.

5.1.1 Experimental Results

We evaluated several pruning methods on **Single Qubit NN**, testing different pruning levels and strategies (local/global, structured/unstructured). The methods were assessed for model accuracy and memory efficiency. The results show that:

- **GP-WMC (Global Pruning with Weight Magnitude Criterion)** achieved the best balance between model size reduction and accuracy retention, particularly at a pruning level of 0.2.

- **Local Unstructured Pruning (LUP)** with weight and gradient magnitude criteria maintained high accuracy, with precise control over sparsity within each layer.

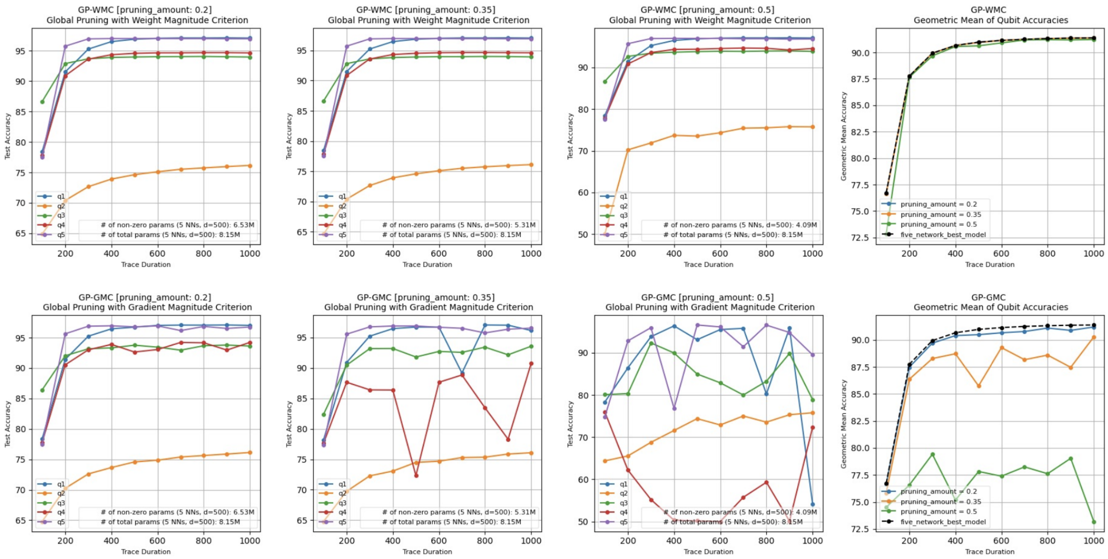

Figure 5.1: Visualization of the Accuracy of Five Individual Qubit Classifiers Across Different Trace Durations 0.2μs to 2μs, and the Geometric Mean Accuracy of a Five Neural Network System Pruned Using Global Pruning with Weight Magnitude Criterion (GP-WMC) and Global Pruning with Gradient Magnitude Criterion (GP-GMC) at Pruning Levels of 0.2, 0.35, and 0.5.

Testing confirmed that GP-WMC at a 0.2 pruning level provided the best balance between model compression and accuracy for qubit readout. Structured Pruning methods effectively reduced model size demonstrated flexibility in sparsity while maintaining robust performance, but exhibited slight accuracy drops at higher pruning levels. Overall, these results demonstrate that effective pruning strategies can yield compact, high-performance models suitable for accurate qubit discrimination in resource-limited environments.

5.2 Quantization

Quantization reduces the size of neural networks by converting floating-point weights and activations to lower-bit integer values, making them more efficient for hardware with limited resources. Dynamic quantization converts weights to int8 during inference, enhancing computational efficiency with minimal accuracy loss. Static quantization applies to both weights and activations and requires calibration with representative data to maintain accuracy. Quantization-aware training (QAT) introduces quantization noise during training, allowing the network to adapt and preserve accuracy, though it requires

Models	Q1	Q2	Q3	Q4	Q5	GM	NZP-1	NZP-5
Single Qubit NN	0.969	0.746	0.940	0.946	0.970	0.910	1.63 M	8.15 M
GP-WMC [0.2] NN	0.969	0.746	0.940	0.946	0.970	0.910	1.3 M	6.53 M
GP-WMC [0.35] NN	0.969	0.746	0.939	0.945	0.970	0.909	1.1 M	5.31 M

Table 5.1: Performance of Individual Qubit Classifiers with Network Pruning Applied at a Trace Duration of 1 μs. In the Table, GP-WMC [0.2] NN and GP-WMC [0.35] NN refer to the models pruned using the Global Pruning with Weight Magnitude Criterion, with pruning amounts of 0.2 and 0.35, respectively. NZP-1 and NZP-5 denote the number of non-zero parameters in a single neural network (1 NN) and a system of five neural networks (5 NN), respectively.

more computational resources. These methods help reduce memory and computational demands, making neural networks suitable for hardware-constrained environments.

5.2.1 Experimental Results

To optimize the **Single Qubit NN** for efficient qubit readout, I applied quantization techniques to reduce the model's memory and computational load while maintaining accuracy.

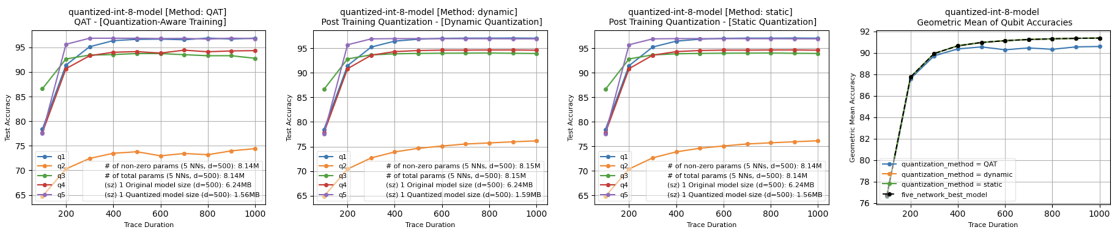

Figure 5.2: Visualization of the Accuracy of Five Individual Qubit Classifiers Across Different Trace Durations 0.2μs to 2μs, and the Geometric Mean Accuracy of a Five Neural Network System Quantized Using Quantization-Aware Training, Post-Training Dynamic Quantization, and Post-Training Static Quantization Methods.

Quantization Methods

Before applying quantization, I performed layer fusion to streamline operations and enhance efficiency. The linear layers were fused with their corresponding batch normalization layers. This actually is significant in terms of reducing the number of operations and improved inference speed. After layer fusion, I applied three quantization methods:

- **Static Quantization**: Weights and activations were quantized to int8, using a calibration phase to set the scaling factors, leading to substantial memory savings with reliable accuracy.
- **Dynamic Quantization**: Weights were converted to int8 at inference, providing flexibility and rapid setup.
- **Quantization-Aware Training (QAT)**: I incorporated QAT during training, enabling the model to adapt to quantization noise, which preserved accuracy better than post-training quantization alone.

Models	Q1	Q2	Q3	Q4	Q5	GM	Size (1 NN)	Size (5 NNs)
Single Qubit NN	0.969	0.746	0.940	0.946	0.970	0.910	6.24 MB	32 MB
[DY] Quantized NN	0.969	0.746	0.940	0.946	0.970	0.910	1.59 MB	8 MB
[ST] Quantized NN	0.969	0.746	0.940	0.946	0.970	0.910	1.56 MB	7.8 MB
[QAT] NN	0.966	0.738	0.937	0.941	0.968	0.906	1.56 MB	7.8 MB

Table 5.2: Performance of Quantized Single Qubit NNs applied at 1μ Trace Duration. In the Table, [DY] Quantized NN, [ST] Quantized NN, and [QAT] NN refer to the models quantized using post-training dynamic quantization, post-training static quantization, and quantization-aware training methods, respectively. Size (1 NN) and Size (5 NNs) indicate the storage size of a single quantized neural network and a system of five quantized neural networks, respectively

In the quantization experiments for single qubit discriminators, I tested dynamic quantization, static quantization, and QAT to evaluate their impact on accuracy and memory efficiency. Across both trace durations of 1μs and 2μs, the post-training models achieved high accuracy but had larger memory footprints. Dynamic quantization significantly reduced model size while maintaining geometric mean (GM) accuracy. Static quantization performed similarly, with even smaller model sizes, making it the most efficient for compression without sacrificing accuracy. Although QAT preserved accuracy during training, it showed a slight drop in GM accuracy compared to the other methods. Consequently, static quantization was identified as the most effective approach for qubit readout, offering an optimal balance of model compression and accuracy retention for resource-constrained applications.

5.3 Knowledge Distillation

Knowledge distillation is a compression technique where a smaller, simpler network, the *student*, learns to replicate the behavior of a larger, complex network, the *teacher*. Introduced by Hinton et al. (2015) [8], knowledge distillation allows the student network to learn from the soft predictions of the teacher, which capture detailed class relationships beyond what hard labels can convey [8].

The knowledge distillation objective minimizes a loss function combining the teacher and student outputs:

$$L_{\text{KD}} = \alpha \cdot L_{\text{CE}}(y, y_{\text{student}}) + (1 - \alpha) \cdot T^2 \cdot L_{\text{KL}}(y_{\text{teacher}}, y_{\text{student}})$$

where L_{CE} is cross-entropy between student predictions and true labels, L_{KL} measures the divergence between teacher and student logits, T is a temperature parameter, and α balances the contributions of the two terms [12].

5.3.1 Knowledge Distillation Architecture

For our knowledge distillation setup, we use a ResNet model as the *teacher* and a lightweight feedforward neural network (FNN) as the *student*. This setup distills complex, high-fidelity predictions from the teacher to a more efficient student model.

Teacher Network: ResNet Architecture

The teacher ResNet model uses residual blocks to capture detailed features, enhancing performance in high-dimensional tasks:

- **Residual Blocks**: Each block consists of three convolutional layers (kernels 8, 5, and 3), followed by batch normalization, ReLU, and 0.5 dropout to improve stability and reduce overfitting [13].
- **Shortcut Connections**: Shortcut connections help prevent vanishing gradients, with a 1x1 convolution aligning dimensions if necessary.
- **Global Average Pooling and Fully Connected Layer**: Averages feature maps to reduce parameters, followed by a final layer for classification.

Experimental results show, this depth and design make the ResNet model suitable for generating high-quality predictions [14].

Student Network: Lightweight FNN

The student model is a compact feedforward neural network, achieving efficient performance through its architecture:

- **Input Layer**: Takes in the same feature inputs as the teacher model.
- **Hidden Layers**: Two hidden layers with 64 and 32 neurons, respectively, each followed by batch normalization, 0.3 dropout, and ReLU activation.
- **Output Layer**: Outputs binary classification predictions.

This compact FNN provides fast inference while maintaining high accuracy through the distilled knowledge from the teacher.

Loss Function: Mean Squared Error (MSE) for Knowledge Distillation

We employed Mean Squared Error (MSE) for knowledge distillation instead of Kullback-Leibler (KL) divergence, due to our use of `BCEWithLogitsLoss` for binary classification. Unlike KL divergence, MSE is appropriate for comparing logits directly, making it computationally simpler and effective in setups where probability distributions are not essential [15]. The MSE loss L_{MSE} is computed as:

$$L_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^{N} (y_{\text{teacher},i} - y_{\text{student},i})^2$$

where N is the number of samples, providing a robust metric for this configuration.

5.3.2 Experimental Results

The results in the tables demonstrate that the Student FNN, distilled from the Teacher Model, achieves comparable accuracy to both the original Single Qubit FNN and the Teacher Model, with minimal loss in performance. For trace duration of 1μs, the Student FNN reaches a GM of 0.910, slightly exceeding the Single Qubit FNN's GM of 0.909, and for trace duration of 2μs, it achieves a GM of 0.914, slightly surpassing both the Single Qubit FNN and Teacher Model (0.913).

Distillation	Q1	Q2	Q3	Q4	Q5	GM	params (1 NN)	Size in (MB)
Single Qubit NN	0.969	0.746	0.940	0.946	0.970	0.910	1.63 M	6.3 MB
Teacher Model	0.969	0.745	0.939	0.945	0.969	0.909	6.9 M	28 MB
Student Model	0.969	0.748	0.940	0.946	0.970	0.910	0.07 M	0.25 MB

Table 5.3: Knowledge Distillation Results for Individual Qubit Classifiers Using the Teacher Model (ResNet), Student Model (Lightweight FNN), and Single Qubit Neural Network for a Trace Duration of 1μs.

This slight improvement in GM for the Student FNN may be attributed to the distillation process acting as a form of regularization. By learning from the Teacher Model's outputs rather than directly from data labels, the Student FNN potentially generalizes better, as the Teacher Model's outputs smooth out noise and inconsistencies in the training data. This effect can lead to marginally improved performance, as seen in the Student FNN's GM values.

Additionally, the Student FNN's compact size, requiring only 0.07 M parameters at trace duration 1 μs and 0.13 M at trace duation 2 μs, highlights the effectiveness of knowledge distillation. Despite its smaller footprint, it maintains accuracy close to the original models, making it highly suitable for deployment on resource-constrained systems.

Distillation	Q1	Q2	Q3	Q4	Q5	GM	params (1 NN)	Size in (MB)
Single Qubit NN	0.971	0.761	0.939	0.946	0.969	0.913	6.5 M	25 MB
Teacher Model	0.971	0.760	0.938	0.946	0.969	0.913	11 M	45 MB
Student Model	0.971	0.764	0.939	0.946	0.970	0.914	0.13 M	0.5 MB

Table 5.4: Knowledge Distillation Results for Individual Qubit Classifiers Using the Teacher Model (ResNet), Student Model (Lightweight FNN), and Single Qubit Neural Network for a Trace Duration of 2μs.

5.4 Network Compression in Chain

To evaluate the impact of sequential compression techniques on qubit readout performance, we applied multiple compression methods to a student feedforward neural network (FNN) model, distilled from a larger teacher model. This approach aimed to maximize compression efficiency while retaining accuracy, making the model suitable for real-time qubit discrimination in resource-constrained environments.

5.4.1 Distillation with Pruning and Quantization

The compression process began with knowledge distillation, where the compact *Student* feedforward neural network (FNN) was trained to replicate the behavior of a more complex *Teacher* ResNet model. By transferring essential knowledge from the teacher model, the student network achieved high accuracy while significantly reducing computational demands, laying a solid foundation for further compression. Following distillation, network pruning was applied to further reduce the student FNN's complexity. Gradient pruning with a weight magnitude criterion was employed, selectively removing 20% of the model's connections based on their weight importance, while preserving critical pathways to maintain performance. This pruning step effectively minimized parameters with minimal accuracy loss, preparing the model for additional compression. Lastly, quantization was introduced to reduce memory and computational requirements even further.

5.4.2 Experimental Results

The experimental results show that combining knowledge distillation, pruning, and quantization leads to significant compression while maintaining high accuracy. Applying gradient pruning with a weight magnitude criterion (GP-WMC) at a pruning level of 0.2, followed by various quantization techniques (static, dynamic, and QAT), resulted in substantial reductions in model size without compromising performance.

For both trace duration of 1 μs and 2 μs, the compressed student models as a five network system achieved geometric mean (GM) accuracy of 0.910 and 0.914, respectively, across all five qubits.

The stability in accuracy, even after aggressive compression, underscores the efficiency of this compression chain. The final model size for each student FNN is remarkably small—0.07 MB for a trace duration of 1 μs and 0.13 MB for a trace length of 2 μs—representing a dramatic reduction in storage requirements compared to the original model.

Compression NNs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**GM**	**params (1 NN)**	**Size (MB)**
GP-WMC [0.2] + [ST] Quant. FNN	0.969	0.748	0.940	0.946	0.970	0.910	0.07 M	0.07 MB
GP-WMC [0.2] + [DY] Quant. FNN	0.969	0.748	0.940	0.946	0.970	0.910	0.07 M	0.07 MB
GP-WMC [0.2] + [QAT] FNN	0.969	0.748	0.940	0.946	0.970	0.910	0.07 M	0.07 MB

Table 5.5: Network Compression Chain Applied to the Knowledge Distillation Student FNN, Using the Best Performing Pruning Method [GP-WMC] (Global Pruning with Weight Magnitude Criterion, Pruning Amount = 0.2), Followed by Static [ST], Dynamic [DY], and Quantization-Aware Training [QAT] Quantization Methods for a Trace Duration of 1 μs.

Compression NNs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**GM**	**params (1 NN)**	**Size (MB)**
GP-WMC [0.2] + [ST] Quant. FNN	0.971	0.764	0.939	0.946	0.970	0.914	0.13 M	0.13 MB
GP-WMC [0.2] + [DY] Quant. FNN	0.971	0.764	0.939	0.946	0.970	0.914	0.13 M	0.13 MB
GP-WMC [0.2] + [QAT] FNN	0.971	0.764	0.939	0.946	0.970	0.914	0.13 M	0.13 MB

Table 5.6: Network Compression Chain Applied to the Knowledge Distillation Student FNN, Using the Best Performing Pruning Method [GP-WMC] (Global Pruning with Weight Magnitude Criterion, Pruning Amount = 0.2), Followed by Static [ST], Dynamic [DY], and Quantization-Aware Training [QAT] Quantization Methods for a Trace Duration of 2 μs.

The uniform performance across static, dynamic, and QAT quantization techniques demonstrates that the compression process preserves accuracy consistently, making this approach flexible and robust. These results highlight the potential of the compression chain as a powerful method to optimize neural networks for qubit readout, achieving a balance of model size reduction and high accuracy that is ideal for practical quantum computing applications.

6 Conclusion and Fututre Work

In this research, we explored the potential of using deep learning techniques for independent qubit readout, focusing on reducing latency and synchronization delays in multi-qubit systems. By addressing key research questions, we validated that independent single-qubit discriminators can achieve high-fidelity readout comparable to traditional multi-qubit models, while overcoming the limitations imposed by synchronized data processing. Our approach combined deep learning with advanced network compression techniques such as knowledge distillation, pruning, and quantization, leading to hardware-efficient models that maintain high accuracy. Through extensive experimentation, we demonstrated that our compressed models achieve performance comparable to their uncompressed counterparts, with significant reductions in memory usage and computational complexity, making them suitable for deployment on resource-constrained hardware like FPGA-based systems.

The results show that network compression, when applied thoughtfully, offers a scalable solution for qubit readout in real-time quantum systems. Our work effectively mitigates some of the challenges posed by crosstalk and computational overhead, offering a promising approach for improving the efficiency of qubit discrimination, particularly in large-scale quantum processors.

While this research has successfully demonstrated the viability of network compression techniques for independent qubit readout, there remain several areas for further investigation and improvement. One key challenge encountered was crosstalk, particularly affecting the performance of Qubit 2, which could be mitigated through additional research into advanced crosstalk reduction techniques. Future work could also focus on optimizing the trace length for each qubit individually, as identifying the optimal measurement lengths may improve readout accuracy and model performance.

Furthermore, the search for well-performing teacher models for knowledge distillation proved challenging, as finding an ideal model that balances complexity with efficiency is critical to the distillation process. Addressing this challenge through more sophisticated model selection or training methods could enhance the performance of the student models even further.

In addition, testing our methods on different datasets and quantum systems will help assess the generalizability of our approach. This will allow us to refine our models to handle a wider range of quantum hardware architectures and signal environments, further validating their robustness and scalability. These future efforts will be crucial in advancing the development of practical, efficient quantum readout systems capable of supporting fault-tolerant quantum computing.

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